

Intelligent Infrastructure: From Edge Attestation to Authorized Physical Consequence

AFA + EABA | Autonomous mobility, physical AI, and locally bounded execution

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PRE-EXAMINATION | BASELINE NOT FROZEN | NO FORMAL DISPOSITION. Hypothetical systems, timestamps, authority claims and records are teaching examples. No affiliation, participation, approval, technical integration or endorsement by Jeffrey DeCoux, Autonomy Institute, or any other organization is implied.

The question this reference examines

An infrastructure system may authenticate a participant, establish a route, and share permission context. Those events do not, by themselves, establish authority for a particular physical movement at the point and moment of execution.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

The objective is not to replace vehicle autonomy, functional safety, traffic management or statutory authority. It is to identify and examine the evidence-to-authority boundary at which a proposed consequence becomes eligible for locally controlled execution.

Reading map

01 / Scope and claims - 02 / Architecture and trust boundaries - 03 / Hypothetical sequence - 04 / Decision method - 05 / Record schema - 06 / Failure modes - 07 / Eight-image storyboard - 08 / Examination pathway.

01 Scope, terminology, and limits

Separate five claims

Claim	What it may establish
Identity / attestation	An asserted or verified property of a component, software state, credential, or edge event - only within the attestation scheme actually specified.
Capability	The system can technically attempt a movement.
Corridor permission	A route- or corridor-level permission claim with issuer, scope and validity conditions.
Local authority	The applicable local actor, rule, and standing governing the particular movement.
Execution authorization	A bounded determination for this action, location, time and current conditions.

Explicit non-claims

This document does not describe or audit Jeffrey DeCoux's actual edge-attestation protocol. It does not assert that any real deployment is unsafe or lacks controls. It is not a safety case, regulatory determination, operational certificate, cybersecurity assessment or permission to operate.

TA14 invariants

Capability is not authority. Observed is not inferred; inferred is not authorized. Authority context may cross entities, including multiple intermediaries, while authority to execute must be established at the relevant local boundary.

02 Architecture and trust boundaries

Illustrative flow: Edge attestation + corridor claim + observed local conditions + applicable local rules + requester standing → evidence and continuity examination → AFA federated context → EABA local execution-boundary evaluation → disposition → local controller and safety systems → observed outcome.

Layer	Role / boundary
Edge attestation (external)	A claim whose issuer, payload, freshness, trust anchor, and limitations must be identified before reliance.
AFA	Carries admissible context across federation boundaries; does not convert that context into blanket execution permission.
EABA	Examines the proposed action against evidence, authority, standing, place, time, and constraints.
Local control / safety	Independently enforces vehicle, intersection, and functional-safety behavior. Governance HOLD is not a braking command.
Outcome evidence	Records what actually happened, separately from what was requested or predicted.

Cross-boundary rule

An upstream claim is not a local mandate. Local authority must be demonstrated, not inferred from a credential, API response, previously commissioned configuration, or corridor allocation.

Where a conflict appears

When a new local restriction conflicts with a still-valid corridor permission, preserve both records and their provenance. Do not silently overwrite either. The local evaluation must identify the controlling rule and the party entitled to apply it.

03 Worked hypothetical: permission becomes stale

Scenario: An autonomous ground vehicle approaches a controlled crossing within an illustrative intelligent infrastructure district. A corridor-level permission exists, but a pedestrian enters the crossing and a local restriction is issued before the proposed movement.

Time	Event	Evidence / authority question
T0 14:00:00	Corridor permission issued for a route and bounded window.	Who issued it, under what scope and revocation rules?
T1 14:00:06	Edge component provides an attestation claim.	Which state was attested, by whom, and how fresh?
T2 14:00:11	Pedestrian enters; local restriction changes.	What is directly observed versus inferred? Which local authority controls?
T3 14:00:12	Movement is proposed at the crossing.	Does prior permission remain sufficient for this exact action NOW?
T4 14:00:12	Illustrative governance disposition: HOLD.	Record the basis; do not treat HOLD as an actuator instruction.
T5 14:00:13	Authorized local controller prevents crossing under its own controls.	What controller log and physical observation confirm the result?
T6 14:00:20	Restriction clears; fresh evidence and authority are examined.	A new ALLOW requires new evaluation, not automatic restoration.

Interpretation: The timestamps and events above are invented. The pedagogical result is HOLD because this example deliberately withholds a sufficient renewed authority basis at T4. It is not a disposition about any real system.

04 Examination logic and dispositions

Preconditions to evaluate

1. Define the proposed consequence precisely: asset, movement, crossing, time and duration. 2. Establish provenance and freshness of relevant observations. 3. Identify applicable rules and competent authority. 4. Establish standing of the requester and issuer. 5. Test whether constraints and revocations have changed. 6. Bind a reasoned disposition to this bounded proposal.

Disposition	Illustrative interpretation
ALLOW	Sufficient evidence, authority and standing exist for the specifically bounded proposed consequence. Not a substitute for controller safety checks.
HOLD	Evidence, authority or standing is unresolved or temporarily insufficient; do not treat prior permission as sufficient.
DENY	A controlling constraint or established disqualifier rules out the proposed consequence.
ESCALATE	Competent authority, conflict resolution or exceptional review is required before a determination.

Safety separation

TA14 describes a governance disposition. The vehicle and infrastructure safety systems retain responsibility for physical motion, obstacle avoidance, emergency stop, and compliance with their own design and regulatory obligations. A governance ALLOW must never override a safety interlock.

Expiry and revocation

An authority decision should be bound to a defined validity window, geographic boundary, action scope, versioned rules, and revocation or invalidation conditions. A change in controlling conditions triggers renewed evaluation where required by the governing rule.

05 Minimum evidence and decision record

Field group	Minimum fields to preserve
Proposal	Action ID, asset, operator, origin/destination, geofence, intended time, exact requested movement.
Attestation	Claim ID, issuer, subject, trust chain, signed payload reference, scope, issuance, expiry, verification result.
Observation	Raw sensor/event record, collection time, source, calibration/status where relevant, uncertainty, provenance.
Authority	Applicable rule, version, jurisdiction, competent issuer, delegation path, standing, constraints, revocations.
Evaluation	Inputs and versions used, admissibility checks, conflicts, rationale, decision maker, disposition and time.
Execution	Local controller acknowledgement, relevant enforcement log, safety interlock state, exception trail.
Outcome	Independent post-event observation, actual crossing/non-crossing, timestamp, discrepancies and follow-up.

Preserve distinctions

A request is not a decision. A decision is not enforcement. An enforcement acknowledgement is not proof of physical outcome. Preserve raw and derived records separately, with the method and uncertainty of any transformation.

Example bounded record (illustrative)

Proposal: VEH-017 / CROSSING-C4 / 14:00:12-14:00:15. Attestation: EDGE-ATT-92 / verified for its stated claim only. Corridor claim: ROUTE-55 / still within stated time window. Local restriction: LOC-RESTR-08 / active. Standing: requesting operator recorded, local override authority not established. Disposition: HOLD / conflict unresolved. Execution evidence: local controller log references inhibited crossing. Outcome: independent observation required to establish actual non-crossing.

06 Pressure tests and failure modes

Challenge	What the examination must ask
Stale-but-valid credential	Can a cryptographically valid attestation still be too old for the proposed physical action?
Conflicting authorities	Which actor is competent at this location, and how is precedence evidenced rather than assumed?
Delayed revocation	What happens if a permission is revoked but the revocation has not reached the edge?
Missing observations	Is an inferred clear crossing being mistaken for observed clear conditions?
No enforcement evidence	Can the operator prove that a HOLD was respected by local control?
Overbroad ALLOW	Is a permission for a district being misapplied to a specific crossing or time?
Restart after HOLD	What fresh evidence and new binding decision are required before resuming?

Adversarial challenge for the architect

Supply one real claim schema and a redacted event chain. Identify what the attestation proves, who can revoke corridor permission, which actor governs the final crossing, and what logs establish physical enforcement. The goal is to find the actual boundary - not to presume a gap.

07 Eight-image showroom production specification

01 THE INTELLIGENT DISTRICT

Visual: Aerial cutaway: autonomous vehicle, intersection, edge node, corridor and separate jurisdictions.

Samantha teaching objective: Teach the difference between shared infrastructure and local operational authority.

02 EDGE ATTESTATION

Visual: Close view of attestation artifact with issuer, subject, time, scope and verified claim.

Samantha teaching objective: An attested property is not execution permission.

03 FEDERATED CONTEXT

Visual: Two jurisdictions and an intermediary passing bounded evidence context; local gate remains distinct.

Samantha teaching objective: AFA transfers context, not automatic authority to act.

04 CONDITIONS CHANGE

Visual: Same crossing, pedestrian entering and new local restriction; show T0 versus T2.

Samantha teaching objective: Previously sufficient evidence can become insufficient.

05 LOCAL EXECUTION BOUNDARY

Visual: Vehicle at crossing with evidence, authority and standing evaluated against current conditions.

Samantha teaching objective: EABA binds the proposed consequence to place and time.

06 HOLD DECISION

Visual: Decision panel: ALLOW / HOLD / DENY / ESCALATE with HOLD selected and reason trace.

Samantha teaching objective: A disposition must be explainable and auditable.

07 LOCAL SAFETY ENFORCEMENT

Visual: Local controller and vehicle safety system maintaining safe behavior; separate governance panel.

Samantha teaching objective: Governance decision is not direct physical actuation.

08 OBSERVED OUTCOME

Visual: Split evidence ledger: request, decision, controller log, independent crossing observation.

Samantha teaching objective: Verify what happened; do not equate logs with physical truth.

Production rules: one illustration per section; numbered teaching labels; no Samantha avatar inside imagery; narration keyed to image number; user-controlled scrolling; educational/hypothetical labels remain visible.

08 Engagement and examination pathway

Step 1 / Independent public demonstration

Publish a clearly independent eight-image teaching showroom and this reference. Do not imply technical collaboration or endorsement from Jeffrey DeCoux or Autonomy Institute.

Step 2 / Written technical exchange

Invite the architect to provide a bounded attestation-to-action sequence: claim semantics, issuer and trust model, permission lifecycle, local authority source, revocation behavior, controller evidence and observed outcome.

Step 3 / Formal examination threshold

A formal TA14 examination requires a registered governance requester with established standing, defined scope, frozen baseline, admissible source records, and agreed evaluation criteria. No real-world ALLOW / HOLD / DENY / ESCALATE is issued in this reference.

TA14 continuity chain

Reality > Record > Continuity > Admissibility > Binding > Commit > Execution > Outcome

No admissible evidence. No admissible execution. Commissioned once does not mean authorized forever.

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